

AEROSPACE ORBITAL GIS CONSOLE

PRAHARI Orbital Console

Real-time spatial intelligence for cyclone tracking & disaster risk — from a photorealistic 3D Earth globe down to sub-meter rooftops, driven by actual project data.

THE PROBLEM

Disaster response is blind at the **last mile**.



Storm data is siloed

Best-track cyclone data, surge models, and ward vulnerability live in separate tools no responder can see together.



Maps are too blurry

Static textures stop at the district level — you can't see which access road, shelter, or rooftop is actually flooded.



Alerts don't reach people

Advisories rarely arrive in the local language or in an offline-ready form for the citizens in the impact zone.

Cyclone Fani (2019) hit Puri at **185 km/h, 932 hPa** — the kind of event where minutes and meters decide outcomes.

OUR SOLUTION

One console. Orbit to rooftop. Real data.

A single web console that fuses live satellite imagery, real cyclone tracks, hydrodynamic surge zones, and infrastructure status into one continuously-zoomable spatial picture — with trilingual citizen advisories.



Seamless multi-scale zoom

3D globe in deep space → regional cyclone swirl → district wards → sub-meter street tiles, with no blurriness.



Live telemetry deck

Central pressure, peak surge, ward vulnerability rankings, and SHAP explainability — updated as you navigate.



Storm time-scrubber

Animate the full Fani best-track from T-48h to landfall to see risk evolve in real time.



Trilingual advisories

Official dispatch orders & 160-char offline SMS templates in Odia, Hindi & English.

SIGNATURE CAPABILITY

Truly zoomable — globe to sub-meter

z = 1–4 · Orbital altitude

Photorealistic 3D Earth globe spinning in black space with atmospheric glow and real orbital tracks.

z = 5–9 · Regional

Atmospheric descent into India & the Bay of Bengal showing the full cyclonic swirl of Fani.

z = 10–13 · District

Puri coast, Chilika Lake, and the 5 administrative wards with real risk tiers.

z = 14–18+ · Sub-meter

Razor-sharp ESRI World Imagery — inspect individual houses, shelters & evacuation corridors.

Basemaps:  ESRI World Imagery ·  NASA GIBS MODIS ·  OpenStreetMap

Talk to your data — the **Gemini AI Advisor**

A built-in agent (Gemini Flash) sits on the right of the console. Municipal & disaster-management officers ask questions in plain language and get grounded, action-oriented answers — no dashboards to decode.



Grounded, never guessing

Every reply is built from the live on-screen data — ward scores, surge depths, shelter status, severed routes. It cites real numbers and says so when data is missing.



Trilingual dispatch

Draft evacuation advisories on demand in Odia, Hindi & English — turning analysis directly into the early-warning message.



Streaming & low-latency

PRAHARI Orbital Console

Answers stream token-by-token with minimal-thinking Gemini Flash, fast enough for a live operations room.



Anticipatory action

"Which wards are most at risk?" · "Are the shelters safe?" · "Which roads are severed?" — decisions in seconds, before landfall.

ZERO MOCKUPS · REAL FIXTURES

Built on **actual** Cyclone Fani data

13

IMD/JTWC best-track waypoints, T-48h → landfall

5

Puri municipal wards scored 33.6–92.4 / 100

3.8 m

Peak storm surge (Holland parametric model)

71%

IoU vs. Sentinel-1 SAR flood ground-truth

3

Multipurpose shelters + severed-corridor status

932 hPa

Landfall central pressure · 185 km/h winds

Surge, pluvial ponding & infrastructure impact from the PRAHARI hazard pipeline — upgradable to live **Google Earth Engine** layers (Sentinel-1 SAR flood, Copernicus DEM surge, WorldPop exposure).

Lightweight, deployable anywhere

Rendering

MapLibre GL JS v5 adaptive globe projection + Three.js for the orbital 3D Earth. No server render needed.

Imagery + GEE

Streamed tiles — ESRI, NASA GIBS, OSM — plus a live **Google Earth Engine** overlay (Sentinel-1 SAR flood, DEM surge) via a tile proxy.

Data layer

Real GeoJSON fixtures (tracks, wards, surge zones, shelters, routes) loaded directly in-browser.

AI Advisor

Gemini Flash called directly from the client, grounded on the live data — no server to run, key stays in the browser.

Zero build step · No backend to operate · Deploys to any static host — GitHub Pages, Netlify, or a field laptop.

WHY IT MATTERS

From satellite data to a **saved life**

- ▶ **Responders** see which shelters are reachable and which access roads are severed — before deploying.
- ▶ **District officials** rank wards by real vulnerability & SHAP drivers to prioritize evacuation.
- ▶ **Citizens** receive clear, trilingual, offline-ready advisories for their exact ward.
- ▶ **Analysts** replay the storm timeline to validate models against SAR ground-truth.

One picture. Orbit to rooftop. In the languages people speak.

THANK YOU

PRAHARI **Orbital Console**

Spatial intelligence & deep-satellite disaster console. Live demo,
working prototype & source available.